

Single-Kernel Convolutional ASIC Accelerator

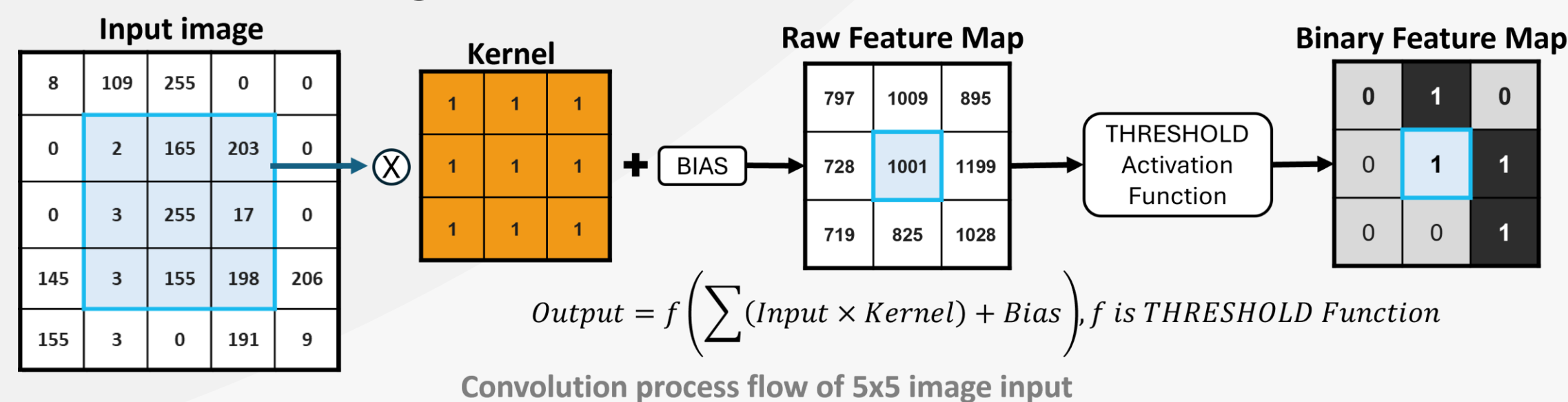
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Introduction

This project presents a custom Area & Power-Optimized ASIC Convolution Accelerator for processing 256x256 grayscale images using a 3x3 kernel. The chip minimizes silicon footprint and power consumption for resource-constrained edge AI devices.



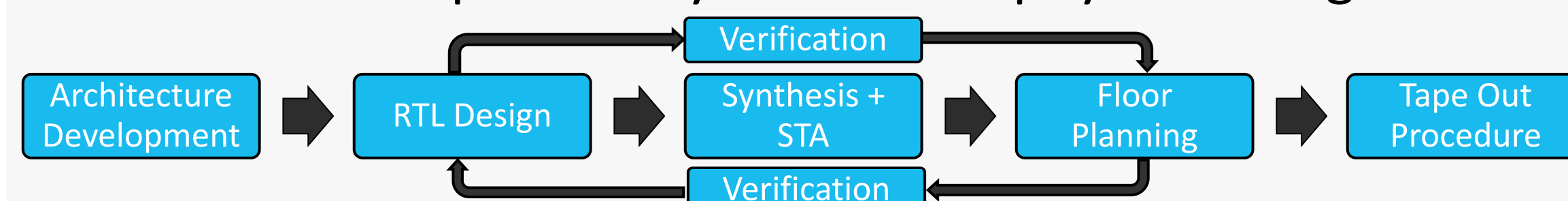
Motivation & Objective

Convolution is the most computationally intensive part of CNNs.

- CPU, GPU, and FPGA-based inference often fail to meet the strict power and area constraints required for energy-limited edge AI devices.
- We propose a standalone ASIC module for a single convolutional kernel as a high-performance, energy-efficient, and area-optimized alternative.
- The design utilizes a full RTL-to-GDSII flow in TSMC 28nm, specifically optimized for minimal area and dynamic power.
- This single convolution unit sets the groundwork for scalable, full-silicon CNN accelerators.

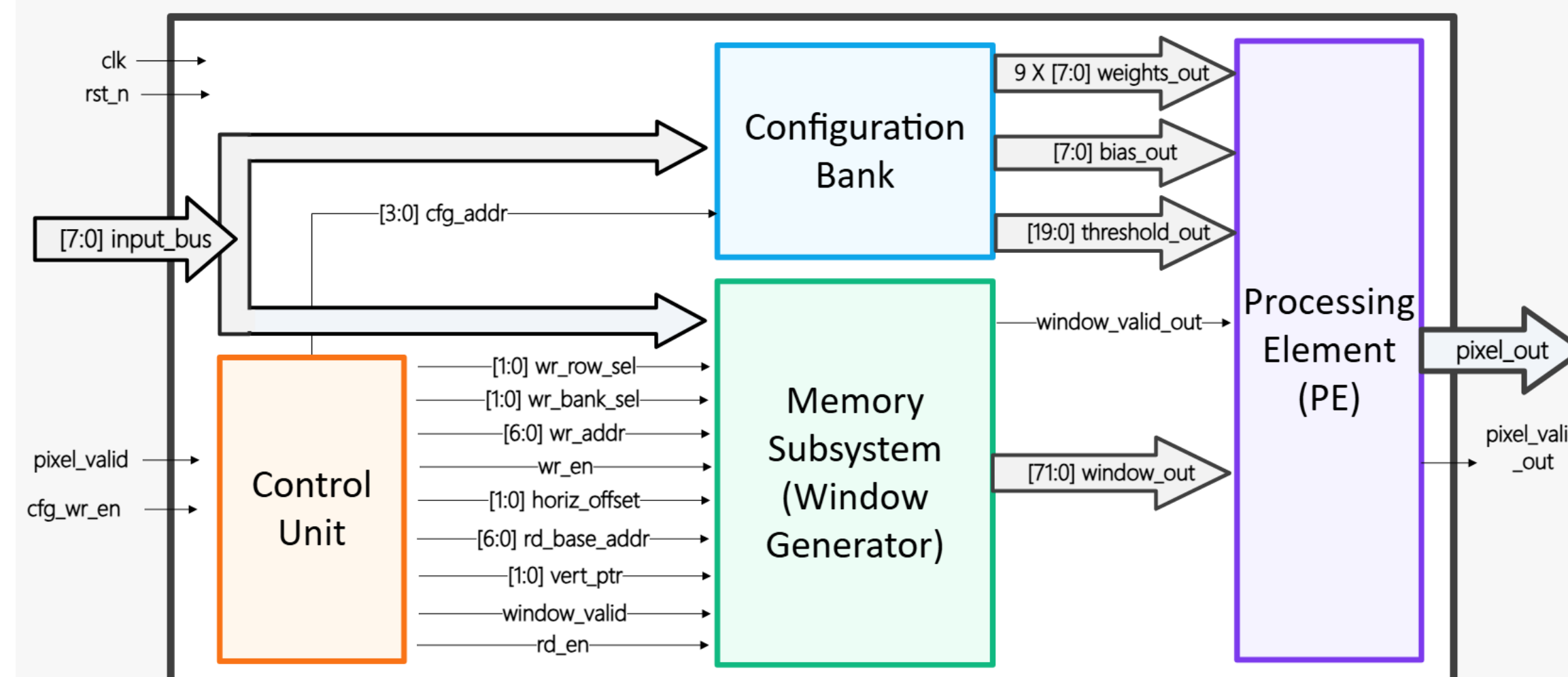
Design Flow

Design implemented using Euclide for RTL and verification, and Fusion Compiler for synthesis and physical design.



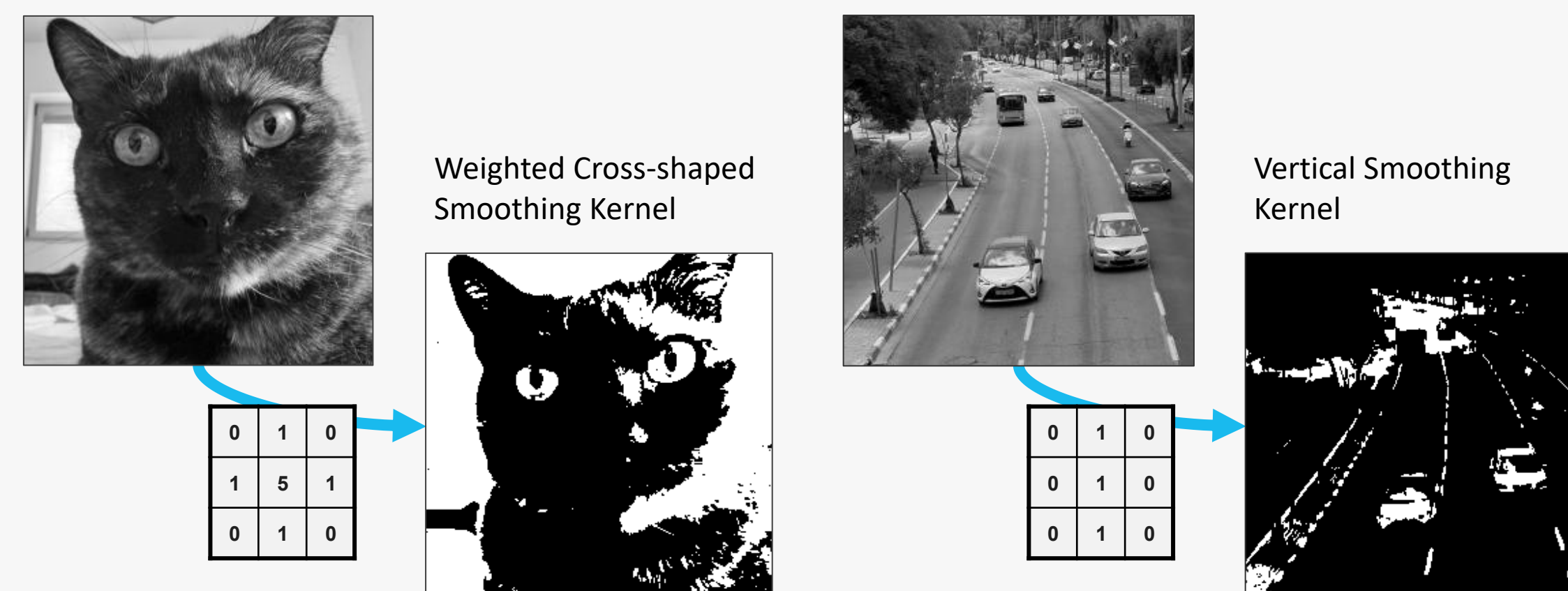
System Architecture

- **Control Unit:** FSM-based orchestrator for dataflow, syncing, and boundary masking.
- **Configuration Bank:** Programmable register array for run-time weight, bias, and threshold updates.
- **Memory Subsystem:** Cyclic 9-bank Interleaved memory enables high-throughput, parallel 3x3 window extraction. Features custom re-ordering logic minimizes area, power, and memory footprint.
- **Processing Element (PE):** Pipelined 9-way MAC engine with adder-tree summation and thresholding.



Results

Post-synthesis verification completed using UVM-based GLS, validating 3x3 convolution accuracy and feature extraction performance.



Physical Implementation

- The physical implementation achieves full timing closure and meets all PPA targets.
- The core encompasses ~22,900 instances (including ~10,800 flip-flops), featuring optimized placement, routing, and clock tree structure.

Target Frequency	400 [MHz]	Timing closure achieved
Worst Slack	+0.80 [ns]	Fully closed and met
Total Dynamic Power	1.70 [mW]	Internal + switching
Leakage Power	23 [μW]	Negligible static power
Core Utilization	62.02%	Site-row based
Core Area	~ 0.0535 [mm^2]	Combinational + sequential logic only
Hold & Open Nets	0	Clean of all routing and timing violations
Clock Gating	1,170 ICGs	Reduced power by 47.5%

At a target frequency of 400 MHz, the accelerator achieves a peak throughput of **~6,100 FPS**, enabling real-time, high-speed image processing with minimal latency.

